Aula 02/04/26

$x = -2 \to \boxed{f(x) = 5x - 1} \to f(x) = -11$ {y, f(x)} | IMAGEM

Domínio

↓

If then else

$f(x) = 5x - 1$

$f(-2) = 5 \cdot (-2) - 1$

$f(-2) = -10 - 1 = -11$

$f(-2) = -11$ $\quad (-2, -11)$

Cálculo da Raiz ⇒ $(X, 0)$

$f(x) = 5x - 1$

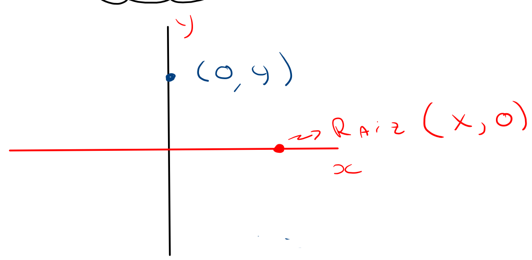
$y = 5x - 1$

$0 = 5x - 1$

$5x - 1 = 0$

$5x = 1$

$x = \frac{1}{5}$ → Raiz $\left(\frac{1}{5}, 0\right)$



Cálculo coeficiente linear

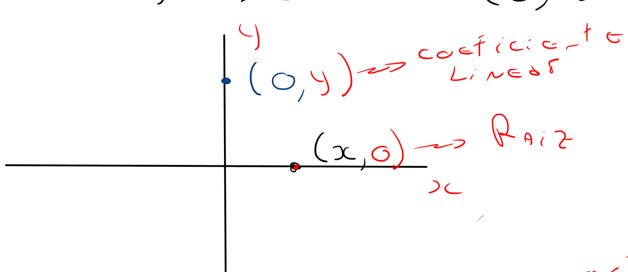
↳ Ponto que corta o Eixo y

$(0, y)$

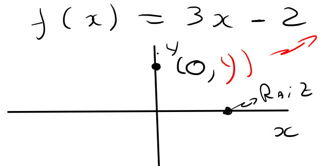
$f(x) = 5x - 1$

$f(0) = 5 \cdot (0) - 1$

$f(0) = -1 \Rightarrow (0, -1)$



$f(x) = 3x - 2$ ⇒ $f(0) = 3 \cdot 0 - 2$, $f(0) = -2$



$(x, 0)$ ⇒ $(2/3, 0)$

$y = 3x - 2$

$0 = 3x - 2$

$3x - 2 = 0$

$x = 2/3$

$f(x) = 3x - 2$

$f(x-1)$

$f(\text{carro}) = 3.(\text{carro}) - 2$

$f(x-1) = 3.(x-1) - 2$

$f(x-1) = 3x - 3 - 2$

$f(x-1) = 3x - 5$

$f(x-1) \longrightarrow \boxed{f(x) = 3x-2} \longrightarrow \quad y = 3x - 5$

$f(x) = x^2 - 4x + 1$

$f(-1) = (-1)^2 - 4.(-1) + 1$

$f(-1) = 1 + 4 + 1 = 6$

$f(-1) = 6 \Rightarrow (-1, 6)$

$f(x) = x^2 - 4x + 1$

$f\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^2 - \frac{4}{1}\left(\frac{1}{2}\right) + 1$

$f\left(\frac{1}{2}\right) = \frac{1}{4} - \frac{4}{2} + \frac{1}{1} = \frac{1 - 8 + 4}{4} = \frac{-3}{4}$

$$\begin{array}{r|l} \text{mmc}\ (2, 4) & 2 \\ (1, 2) & 2 \\ \to (1, 1) & 4 \Rightarrow \text{Denominador} \end{array}$$

$f\left(\frac{1}{2}\right) = \frac{-3}{4} \Rightarrow \left(\frac{1}{2}, -\frac{3}{4}\right)$

Exercícios

1) $f(x) = x^3 - 2x + 1$

p/ $x = -1$

Domínio: $x = -1 \longrightarrow \boxed{f(x) = x^3 - 2x + 1} \longrightarrow f(x) = ?$ Imagem

$f(-1) = (-1)^3 - 2.(-1) + 1$

$f(-1) = -\cancel{1} + 2 + \cancel{1}$

$f(-1) = 2 \Rightarrow (-1, 2)$

$f(x) = \frac{4x^3 - 2x + 1}{3x - 2}$

$f(-2) = \frac{4.(-2)^3 - 2(-2) + 1}{3.(-2) - 2} = \frac{4.(-8) + 4 + 1}{-6 - 2} = \frac{-32 + 4 + 1}{-8} = \frac{-27}{-8} = \frac{27}{8}$